CS 306: Wireless Networks (Part-II)

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Outline

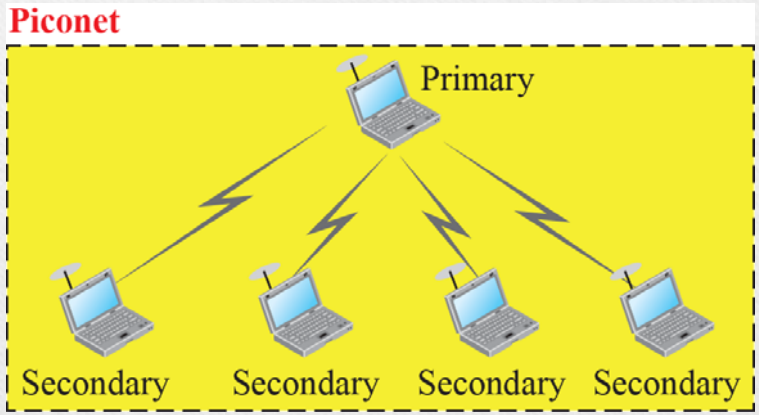
- Bluetooth
- WiMAX

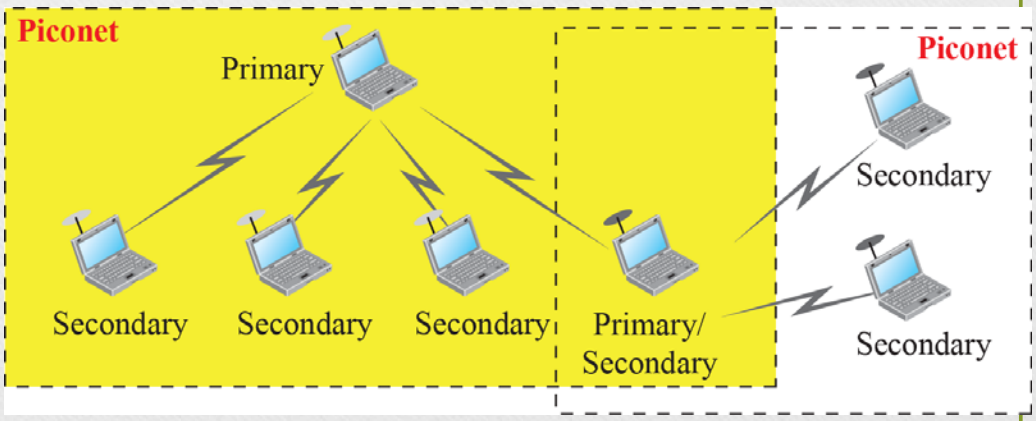
BLUETOOTH

- Bluetooth is a wireless LAN technology designed to connect devices of different functions when they are at a short distance from each other.
- A Bluetooth LAN is an ad hoc network.
- The devices, sometimes called gadgets, find each other and make a network called a piconet.

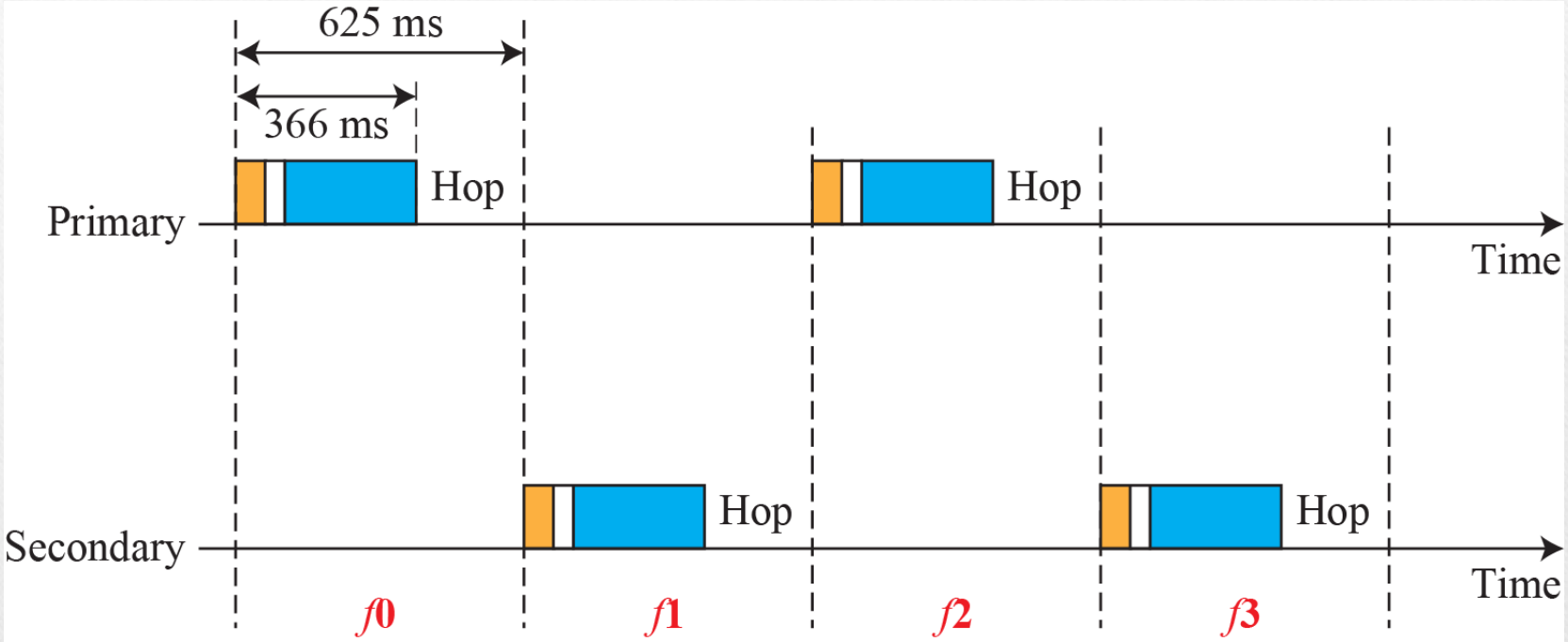
Architecture

- Bluetooth defines two types of networks:
  - Piconet
  - Scatternet

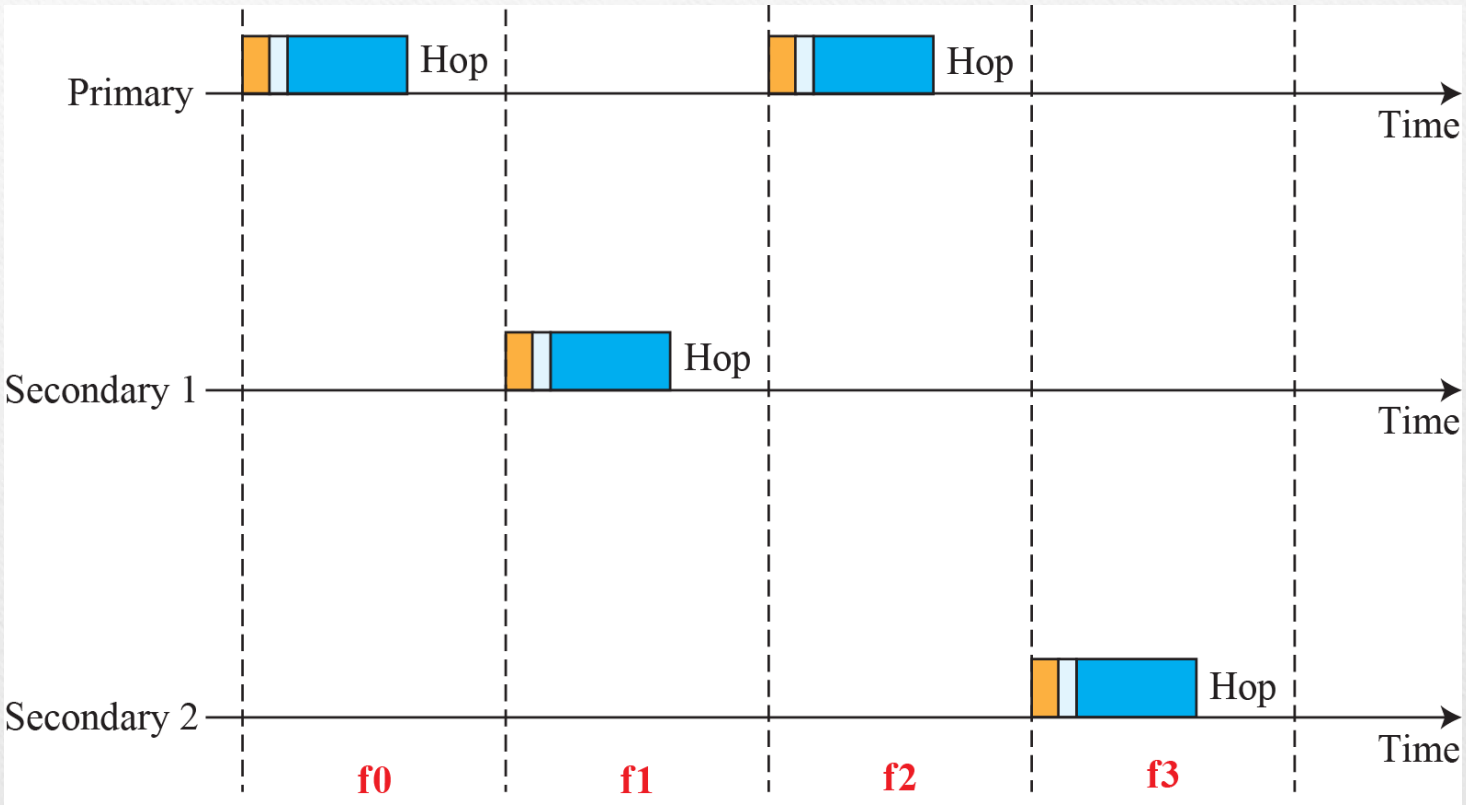




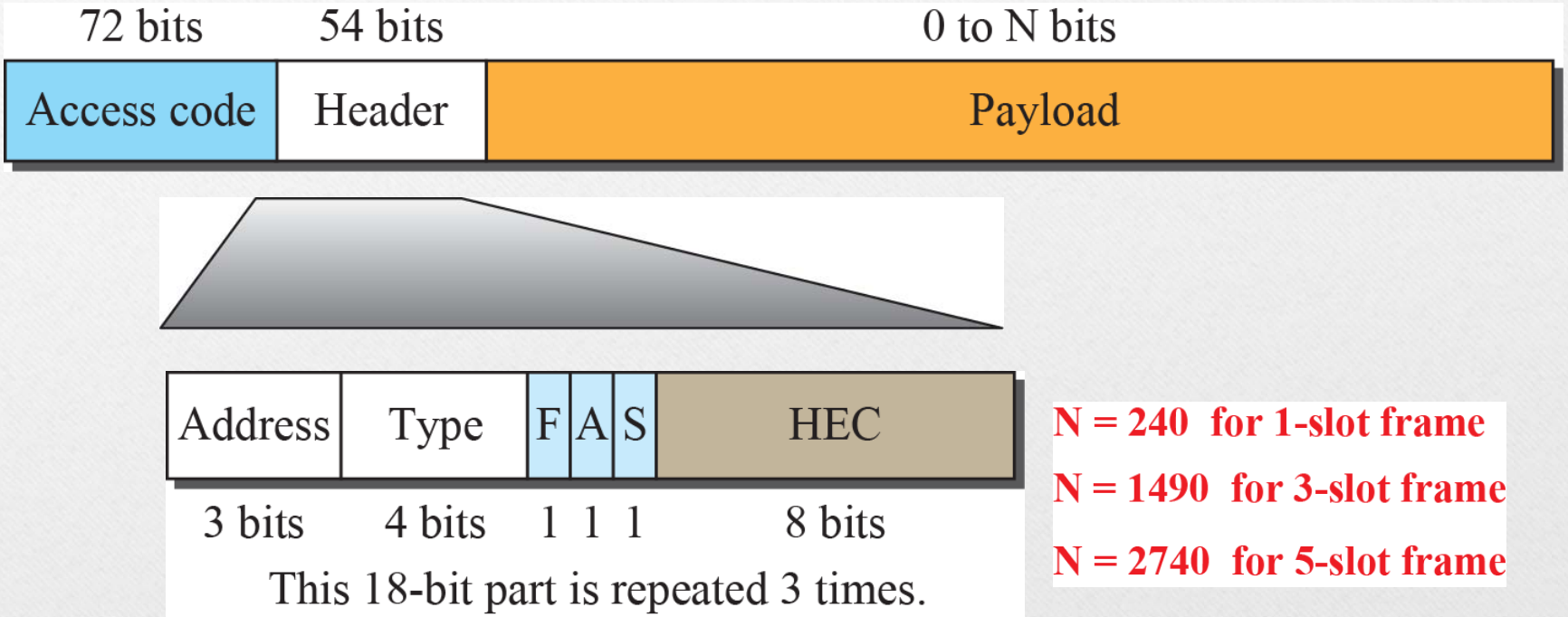
Single-secondary communication



Multiple-secondary communication



Frame format



N = 240 for 1-slot frame

N = 1490 for 3-slot frame

N = 2740 for 5-slot frame

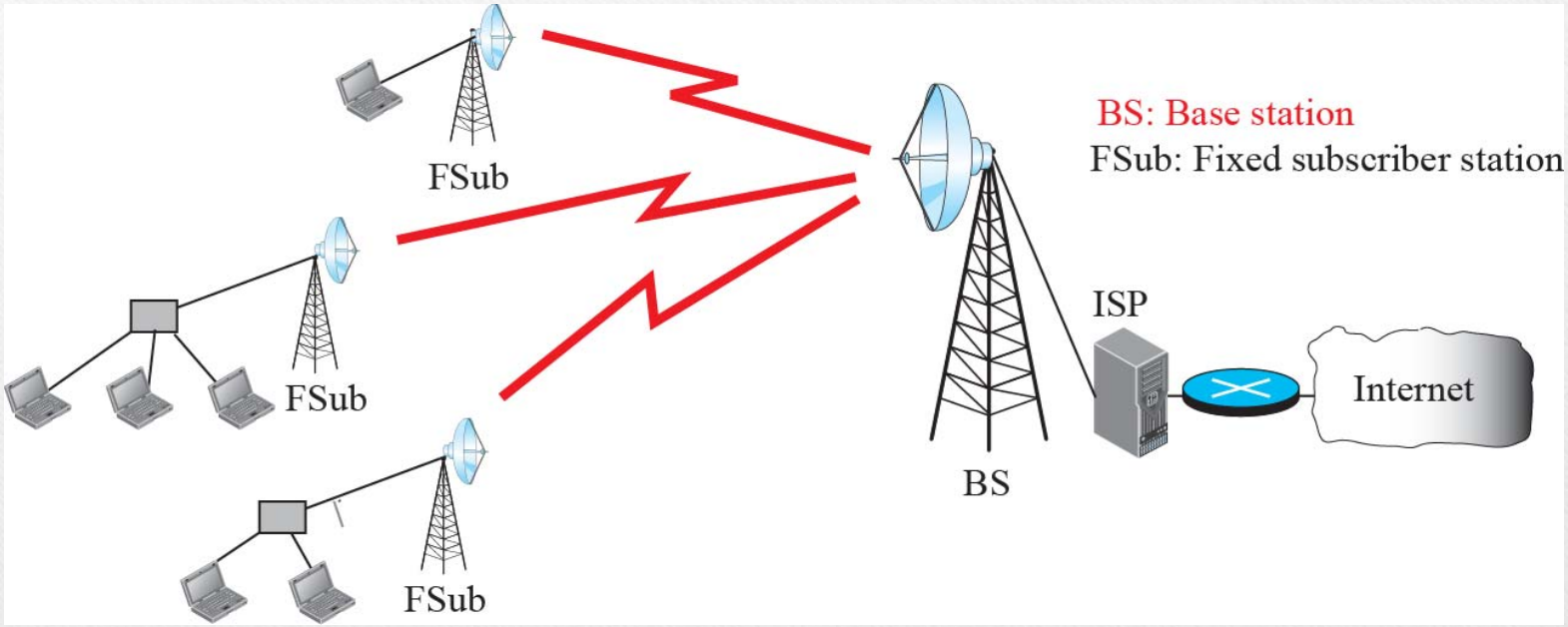
WiMAX

- WiMAX has been designed for the types of applications
  - People want to have access to the Internet from home or office (fixed) where the wired access to the Internet is either not available or is expensive.
  - People also need to access the Internet when they are using their cellular phones.

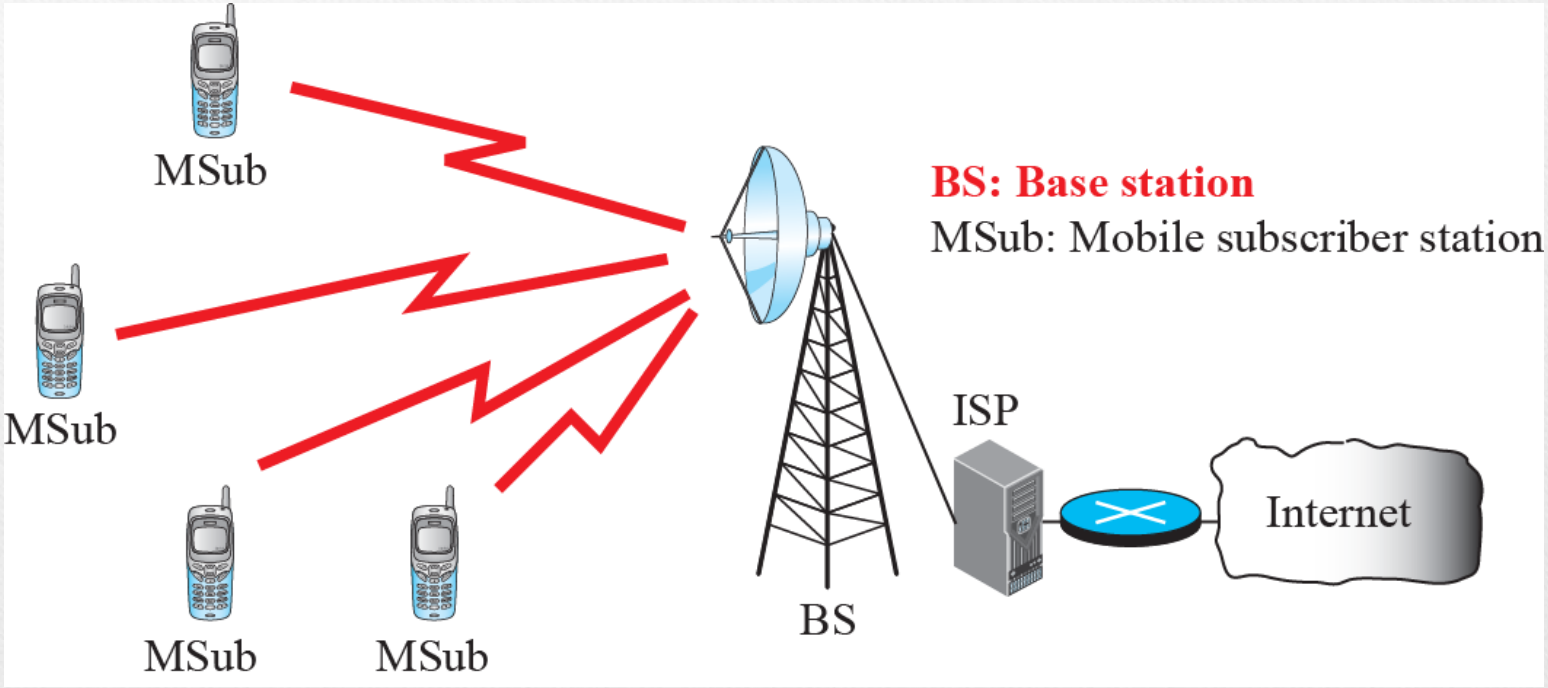
Services

- WiMAX provides two types of services to subscribers:
  - Fixed
  - Mobile

Fixed WiMAX


FSub
FSub
FSub
BS: Base station
FSub: Fixed subscriber station
ISP
Internet
BS

Mobile WiMAX

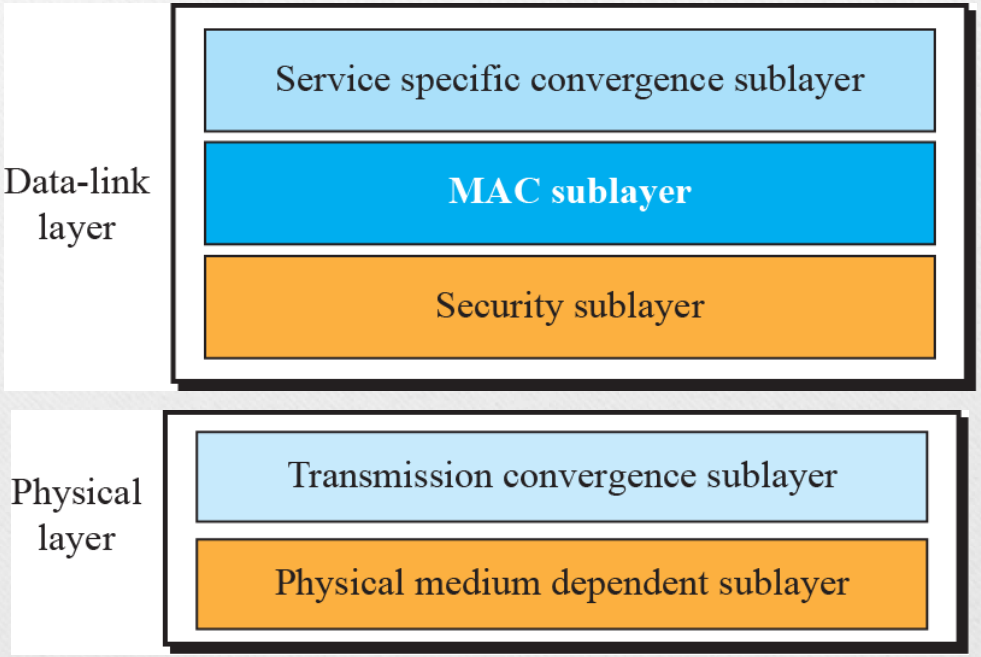


IEEE Project 802.16

- WiMAX is the result of the IEEE 802.16 project.
- The standard is sometimes referred to as wireless local loop, in contrast with wired local loop (dial-up, DLS, or cable).
- IEEE 802.16 is a standard for a wireless WAN (or MAN).
- Project 802.11 defines a connectionless communication; project 802.16 defines a connection-oriented service.

Layers

- IEEE has divided the data-link layer into three sublayers and the physical layer into two sublayers.



Next Class

- Cellular Telephony
- Satellite Networks

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